

Pseudomonas spp.

Expert Rules and Expected Phenotypes

Guidance documents

EUCAST Clinical Breakpoint Tables v. 15.0, valid from 2025-01-01

For abbreviations and explanations of breakpoints, see the Notes sheet

MIC determination (broth microdilution according to ISO standard 20776-1)

Medium: Cation-adjusted Mueller-Hinton broth (for cefiderocol, see

<https://www.eucast.org/eucastguidancedocuments/>

Inoculum: 5x10⁵ CFU/mL

Incubation: Sealed panels, air, 35±1°C, 18±2h

Reading: Unless otherwise stated, read MICs at the lowest concentration of the agent that completely inhibits visible growth. See "EUCAST Reading Guide for broth microdilution" for further information.

Quality control: *Pseudomonas aeruginosa* ATCC 27853. For agents not covered by this strain and for control of the inhibitor component of beta-lactam inhibitor combinations, see EUCAST QC Tables.

Disk diffusion (EUCAST standardised disk diffusion method)

Medium: Mueller-Hinton agar

Inoculum: McFarland 0.5

Incubation: Air, 35±1°C, 18±2h

Reading: Unless otherwise stated, read zone edges as the point showing no growth viewed from the back of the plate against a dark background illuminated with reflected light. See "EUCAST Reading Guide for disk diffusion" for further information.

Quality control: *Pseudomonas aeruginosa* ATCC 27853. For agents not covered by this strain and for control of the inhibitor component of beta-lactam inhibitor-combination disks, see EUCAST QC Tables.

Pseudomonas aeruginosa is the most frequent species of this genus. Other less frequent *Pseudomonas* species recovered in clinical samples are: *P. fluorescens* group, *P. putida* group and *P. stutzeri* group.

Penicillins	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Benzylpenicillin	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method. 1. For susceptibility testing purposes, the concentration of tazobactam is fixed at 4 mg/L. 2. For susceptibility testing purposes, the concentration of clavulanic acid is fixed at 2 mg/L.
Ampicillin	-	-			-	-		
Ampicillin-sulbactam	-	-			-	-		
Amoxicillin	-	-			-	-		
Amoxicillin-clavulanic acid	-	-			-	-		
Piperacillin	0.001	16		30	50	18	18-19	
Piperacillin-tazobactam	0.001 ¹	16 ¹		30-6	50	18	18-19	
Ticarcillin-clavulanic acid	0.001 ²	16 ²		75-10	50	18		
Temocillin	-	-			-	-		
Phenoxymethylpenicillin	-	-			-	-		
Oxacillin	-	-			-	-		
Cloxacillin	-	-			-	-		
Dicloxacillin	-	-			-	-		
Flucloxacillin	-	-			-	-		
Mecillinam oral (pivmecillinam) (uncomplicated UTI only)	-	-			-	-		

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Cephalosporins	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Cefaclor	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method. <u>1/A. The addition of a beta-lactamase inhibitor does not add clinical benefit. The beta-lactamases produced by the organisms either do not modify the parent cephalosporin or are insufficiently inhibited by the inhibitor.</u> 2. Broth microdilution MIC determination must be performed in iron-depleted Mueller-Hinton broth and specific reading instructions must be followed. For testing conditions and reading instructions, see https://www.eucast.org/eucastguidancedocuments/. 3. For susceptibility testing purposes, the concentration of avibactam is fixed at 4 mg/L. 4. See table of dosages for dosing for different indications. 5. For susceptibility testing purposes, the concentration of tazobactam is fixed at 4 mg/L.
Cefadroxil	-	-			-	-		
Cefalexin	-	-			-	-		
Cefazolin	-	-			-	-		
Cefepime	0.001	8		30	50	21	19-23	
Cefepime-enmetazobactam	Note ¹	Note ¹			Note ^A	Note ^A		
Cefiderocol, <i>P. aeruginosa</i>	2 ²	2 ²		30	22	22	20-21	
Cefixime	-	-			-	-		
Cefotaxime	-	-			-	-		
Cefoxitin	-	-			-	-		
Cefpodoxime	-	-			-	-		
Ceftaroline	-	-			-	-		
Ceftazidime	0.001	8		10	50	17		
Ceftazidime-avibactam, <i>P. aeruginosa</i>	8 ³	8 ³		10-4	17	17	16-17	
Ceftibuten	-	-			-	-		
Ceftobiprole	IE	IE			IE	IE		
Ceftolozane-tazobactam ⁴ , <i>P. aeruginosa</i>	4 ⁵	4 ⁵		30-10	23	23		
Ceftriaxone	-	-			-	-		
Cefuroxime iv	-	-			-	-		
Cefuroxime oral	-	-			-	-		

Carbapenems	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Doripenem	0.001	2		10	50	22		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method. 1. For susceptibility testing purposes, the concentration of relebactam is fixed at 4 mg/L. 2. For susceptibility testing purposes, the concentration of vaborbactam is fixed at 8 mg/L.
Ertapenem	-	-			-	-		
Imipenem	0.001	4		10	50	20		
Imipenem-relebactam, <i>P. aeruginosa</i>	2 ¹	2 ¹		10-25	22	22		
Meropenem (indications other than meningitis), <i>P. aeruginosa</i>	2	8		10	20	14		
Meropenem (indications other than meningitis), <i>Pseudomonas</i> other than <i>P. aeruginosa</i>	2	8		10	24	18		
Meropenem (meningitis), <i>P. aeruginosa</i>	2	2		10	20	20		
Meropenem-vaborbactam, <i>P. aeruginosa</i>	8 ²	8 ²		20-10	14	14		

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Monobactams	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Aztreonam	0.001	16		30	50	18		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Aztreonam-avibactam	IE	IE			IE	IE		

Fluoroquinolones	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Ciprofloxacin	0.001	0.5		5	50	26		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Delafloxacin	IE	IE			IE	IE		
Levofloxacin	0.001	2		5	50	18		
Moxifloxacin	-	-			-	-		
Nalidixic acid (screen only)	NA	NA			NA	NA		
Norfloxacin (uncomplicated UTI only)	-	-			-	-		
Ofloxacin	-	-			-	-		

Aminoglycosides ¹	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Amikacin (systemic infections)	(16) ¹	(16) ¹		30	(15) ^A	(15) ^A		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method. 1/A. For information on how to use breakpoints in brackets, see https://www.eucast.org/eucastguidancedocuments/ .
Amikacin (infections originating from the urinary tract)	16	16		30	15	15		
Gentamicin (systemic infections)	IE	IE			IE	IE		
Gentamicin (infections originating from the urinary tract)	IE	IE			IE	IE		
Netilmicin	IE	IE			IE	IE		
Tobramycin (systemic infections)	(2) ¹	(2) ¹		10	(18) ^A	(18) ^A		
Tobramycin (infections originating from the urinary tract)	2	2		10	18	18		

Glycopeptides and lipoglycopeptides	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Dalbavancin	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Oritavancin	-	-			-	-		
Teicoplanin	-	-			-	-		
Telavancin	-	-			-	-		
Vancomycin	-	-			-	-		

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Macrolides, lincosamides and streptogramins	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Azithromycin	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Clarithromycin	-	-			-	-		
Erythromycin	-	-			-	-		
Roxithromycin	-	-			-	-		
Clindamycin	-	-			-	-		
Quinupristin-dalfopristin	-	-			-	-		

Tetracyclines	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Doxycycline	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Eravacycline	-	-			-	-		
Minocycline	-	-			-	-		
Tetracycline	-	-			-	-		
Tigecycline	-	-			-	-		

Oxazolidinones	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Linezolid	-	-			-	-		Numbered notes relate to general comments and/or MIC breakpoints. Lettered notes relate to the disk diffusion method.
Tedizolid	-	-			-	-		

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Miscellaneous agents	MIC breakpoints (mg/L)			Disk content (µg)	Zone diameter breakpoints (mm)			Notes
	S ≤	R >	ATU		S ≥	R <	ATU	
Chloramphenicol	-	-			-	-		1. Colistin MIC determination should be performed with broth microdilution. Quality control must be performed with both a susceptible QC strain (<i>E. coli</i> ATCC 25922 or <i>P. aeruginosa</i> ATCC 27853) and the colistin resistant <i>E. coli</i> NCTC 13846 (<i>mcr-1</i> positive). 2. For information on how to use breakpoints in brackets, see https://www.eucast.org/eucastguidancedocuments/. 3/B. Antimicrobial susceptibility testing is discouraged. For information on the use of fosfomycin iv in combination therapy, see https://www.eucast.org/eucastguidancedocuments/. A. Use an MIC method (broth microdilution only).
Colistin ¹	(4) ²	(4) ²			Note ^A	Note ^A		
Daptomycin	-	-			-	-		
Fosfomycin iv	Note ³	Note ³			Note ^B	Note ^B		
Fosfomycin oral	-	-			-	-		
Fusidic acid	-	-			-	-		
Lefamulin	-	-			-	-		
Metronidazole	-	-			-	-		
Nitrofurantoin (uncomplicated UTI only)	-	-			-	-		
Nitroxoline (uncomplicated UTI only)	-	-			-	-		
Rifampicin	-	-			-	-		
Spectinomycin	-	-			-	-		
Trimethoprim (uncomplicated UTI only)	-	-			-	-		
Trimethoprim-sulfamethoxazole	-	-			-	-		